

Automatic Image Analysis

Questions for exam preparation

Part I

1. What is the difference between structured and unstructured data?
2. What is top-down processing?
3. What is bottom-up processing?

4. What is object recognition?
5. What is object detection?
6. What is object classification?
7. What is semantic segmentation?
8. How does a typical processing chain for semantic image analysis look like?
9. What are typical classifiers used in image analysis tasks?
10. What is the general idea of the “Bag of (Visual) Words” approach?

11. What is a linear classifier?
12. What does MLP stand for?
13. What is the general principle of an MLP?
14. What is usually referred to as “backprop”?
15. What does RF stand for?
16. What is the general principle of an RF?

17. What is the general principle of the Fourier transformation?
18. What is the Fourier spectrum of a sine function?
19. What is the Fourier spectrum of a rectangular function?
20. What is the Fourier spectrum of a sinc function?
21. What is the convolution theorem?
22. What is the correlation theorem?
23. Which property does the spectrum of a real function fulfill?

24. How to extract a normalized Fourier descriptor given a grayscale image?
25. How does the Fourier descriptor change when the object is translated?
26. How does the Fourier descriptor change when the object is scaled?
27. How does the Fourier descriptor change when the object is rotated?
28. How to smooth a shape using the framework of Fourier descriptors?
29. What are disadvantages of Fourier descriptors, i.e. when will they fail?

30. Is biological saliency a pure bottom-up, a pure top-down process, or a mixture of both?
31. What is the aim of saliency operators for image analysis?
32. Describe a computational model of saliency.
33. What is the entropy of a signal?
34. Does a textured image have a higher or lower entropy than a homogeneous image?
35. What is an image pyramid?
36. What is a linear Gaussian scale space?
37. What is the modern approach to image saliency?
38. What are possible applications of saliency operators for images?

39. State a parametric model of a line.
40. How is a point on a line in image space represented in the parameter space, if the line detector is based on the Radon transformation?
41. How is a point on a line in image space represented in the parameter space, if the Hough line detector is used?
42. What is the working principle of the Hough line detector?
43. What is the relationship between the gradient of a line and the orientation of the line?
44. How many dimensions has a Hough circle detector?
45. What is the R-table?
46. How many dimensions has the voting space of the Generalized Hough Transform?
47. What does ISM stand for?
48. What are the differences between the Generalized Hough Transform and the ISM?
49. What is the algorithm to train the ISM?
50. What is the algorithm to apply the ISM?
51. What is a Hough Forest?
52. What kind of node tests are used in a Hough Forest?
53. What kind of energy functions are used in a Hough Forest?
54. Which information is stored in the leafs of a Hough Forest?
55. What does HOG stand for?
56. How is a HOG descriptor computed given an image?
57. What does DPM stand for?
58. What is the general working principle of a DPM?